

Matthew L. Bendall

 Washington, DC
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  Google Scholar

Professional Summary

- **Publications:** 38 peer-reviewed publications, 10 as first or senior author; h-index 18
- **Teaching and mentoring:** formal mentoring of 5 postdoctoral scholars or graduate students; advising and supervision of 11 trainees; four years as Lecturer for Computational Biology Master's Thesis Exploration (WCM); two years as Teaching Assistant for upper-level computational biology courses (BYU); lead organizer and instructor for Dark Matter Codeathon (WCM); instructor at 4 workshops on transposable elements or microbiome
- **Research funding:** Principal Investigator on WCM Fund for the Future grant (completed 2024); Co-Investigator on NCI (R01CA260691) and NIAID (UM1AI164559) (through May 2026); Co-Investigator on NIDA SCORCH grant (through March 2030)
- **Scientific contributions:** lead developer of four open-source software packages, including Telescope (100+ citations); author and developer of Pathoscope (100+ citations); contributor to other open-source bioinformatics projects such as Bioconda (600+ citations)

Education

Ph.D.	George Washington University , Genomics and Bioinformatics	Washington, DC May 2019
M.S.	Brigham Young University , Biology	Provo, UT Aug 2012
B.S.	Brigham Young University , Bioinformatics	Provo, UT Aug 2009
A.A.	Montgomery College , Computer Science	Rockville, MD May 2006

Experience

The George Washington University , Associate Research Professor Department of Biostatistics and Bioinformatics, Milken Institute School of Public Health	Washington, DC Sept 2025 – present 1 year 2 months
Weill Cornell Medicine , Assistant Professor of Computational Biology Research in Medicine Division of Infectious Diseases, Department of Medicine	New York, NY Aug 2019 – Aug 2025 6 years 1 month
Nixon Lab, George Washington University School of Medicine , Doctoral Research Assistant Advisor: Douglas Nixon, MD, PhD	Washington, DC Jan 2015 – July 2018 3 years 7 months
• Developed Telescope, a computational approach for quantifying transposable element expression using RNA-seq. • Investigated HERV expression profiles in cancer, neurodevelopment, psychiatry, and embryonic development.	

Computational Biology Institute, George Washington University Milken Institute of Public Health, Doctoral Research Assistant

Advisor: Keith Crandall, PhD

- Investigated microbiome community composition and host-pathogen interactions in schizophrenia, pediatric asthma, and cystic fibrosis.
- Performed statistical analysis for the WE-HEAL study and identified transcriptional patterns in hidradenitis suppurativa.
- Developed the software package HAPHPIPE for viral haplotype assembly and phylodynamics; characterized HIV phylodynamics in Washington DC.

Washington, DC
Aug 2014 – July 2019
5 years

Joint Genome Institute, Lawrence Berkeley National Laboratory, Bioinformatics Software Developer

Synthetic Biology (Samuel Deutsch, PhD); Single Cell Genomics (Rex Malmstrom, PhD)

- Implemented analysis pipelines for the assembly, mapping, and annotation of metagenomic data using high-performance computing. Developed approaches for detecting selective sweeps using metagenomic time-series.
- Led a team of software developers in building laboratory information management systems to support JGI's Synthetic Biology and Single Cell Genomics programs.
- Developed novel methods for identifying epigenetic modifications using PacBio's single-molecule real-time (SMRT) sequencing technology.

Walnut Creek, CA
Dec 2011 – Aug 2014
2 years 9 months

Crandall Lab, Brigham Young University, Graduate Research Assistant

Advisor: Keith Crandall, PhD

- Developed the software package Pathoscope for strain level quantification of metagenomic NGS data.
- Evaluated phylogenetic node calibration methods using empirical phylogenetic and fossil history.
- Compared phylogenetic and structural approaches for identifying deleterious variants in ANGPTL4.
- Evaluated the performance of methods for predicting sites under positive selection in drug-resistant HIV (thesis).

Provo, UT
Aug 2009 – Dec 2011
2 years 5 months

Crandall Lab, Brigham Young University, Undergraduate Research Assistant

Mentor: Nicole Lewis Rogers, PhD

- Examined phylogenetic relationships among 300+ enterovirus isolates to test hypotheses of recombination and adaptation.

Provo, UT
Oct 2007 – Aug 2009
1 year 11 months

Publications

A single-cell transposable element atlas of human cell identity

July 2025

Reyes-Gopar H, Marston JL, Singh B, Greenig M, Lin J, Ostrowski MA, Randall KN Jr, Sandoval-Motta S, Dopkins N, Lawrence E, O'Mara MM, Fei T, Duarte RRR, Powell TR, Hernández-Lemus E, Iñiguez LP, Nixon DF, Bendall ML

[10.1016/j.crmeth.2025.101086](https://doi.org/10.1016/j.crmeth.2025.101086) (Cell Rep Methods 5(7):101086)

Integration of chromosome conformation and gene expression networks reveals regulatory mechanisms in triple negative breast cancer

Jan 2025

Reyes-Gopar H, Pérez-Fuentes KA, Bendall ML, Hernández-Lemus E

[10.3389/fcell.2025.1597245](https://doi.org/10.3389/fcell.2025.1597245) (Front Cell Dev Biol 13:1597245)

Locus-specific human endogenous retroviruses reveal lymphoma subtypes

June 2025

Singh B, Dopkins N, Fei T, Marston JL, Michael S, Reyes-Gopar H, Curty G, Heymann JJ, Chadburn A, Martin P, Leal FE, Cesarman E, Nixon DF, Bendall ML

[10.1016/j.isci.2025.112541](https://doi.org/10.1016/j.isci.2025.112541) (iScience 28(6):112541)

Transposable elements may enhance antiviral resistance in HIV-1 elite controllers Singh M, Leddy SM, Iñiguez LP, Bendall ML, Nixon DF, Feschotte C 10.1186/s13059-025-03484-y (Genome Biol 26(1):28)	Feb 2025
Endogenous retroelement expression in modeled airway epithelial repair Michael S, Liotta N, Fei T, Bendall ML, Nixon DF, Dopkins N 10.1016/j.micinf.2024.105465 (Microbes Infect)	Dec 2024
Retro-age: A unique epigenetic biomarker of aging captured by DNA methylation states of retroelements Ndhlovu LC, Bendall ML, Dwaraka V, Pang APS, Dopkins N, Carreras N, Smith R, Nixon DF, Corley MJ 10.1111/accel.14288 (Aging Cell 23(10):e14288)	Oct 2024
Integrating human endogenous retroviruses into transcriptome-wide association studies highlights novel risk factors for major psychiatric conditions Duarte RRR, Pain O, Bendall ML, de Mulder Rougvié M, Marston JL, Selvakadunco S, Troakes C, Leung SK, Bamford RA, Mill J, O'Reilly PF, Srivastava DP, Nixon DF, Powell TR 10.1038/s41467-024-48153-z (Nat Commun 15(1):3803)	May 2024
Endogenous retroelement expression in the gut microenvironment of people living with HIV-1 Dopkins N, Fei T, Michael S, Liotta N, Guo K, Mickens KL, Barrett BS, Bendall ML, Dillon SM, Wilson CC, Santiago ML, Nixon DF 10.1016/j.ebiom.2024.105133 (EBioMedicine 103:105133)	May 2024
Ribosomal profiling of human endogenous retroviruses in healthy tissues Dopkins N, Singh B, Michael S, Zhang P, Marston JL, Fei T, Singh M, Feschotte C, Collins N, Bendall ML, Nixon DF 10.1186/s12864-023-09909-x (BMC Genomics 25(1):5)	Jan 2024
Oncogenic Transformation Drives DNA Methylation Loss and Transcriptional Activation at Transposable Element Loci Kanholt T, Rentia U, Hadley M, Karlow JA, Cox OL, Diab N, Bendall ML, Dawson T, McDonald JI, Xie W, Crandall KA, Burns KH, Baylin SB, Easwaran H, Chiappinelli KB 10.1158/0008-5472.CAN-22-3485 (Cancer Res 83(15):2584-2599)	Aug 2023
Endogenous Reverse Transcriptase Inhibition Attenuates TLR5-Mediated Inflammation Dopkins N, Singh B, Michael S, O'Mara MM, Marston JL, Fei T, Bendall ML, Nixon DF 10.1128/mbio.03280-22 (mBio 14(1):e0328022)	Feb 2023
Naso-oro-pharyngeal microbiome from breast cancer patients diagnosed with COVID-19 Viana MC, Curty G, Furtado C, Singh B, Bendall ML, Viola JPB, de Melo AC, Soares MA, Moreira MAM 10.3389/fmicb.2022.1074382 (Front Microbiol 13:1074382)	Jan 2022
A field guide to endogenous retrovirus regulatory networks Dopkins N, O'Mara MM, Lawrence E, Fei T, Sandoval-Motta S, Nixon DF, Bendall ML 10.1016/j.molcel.2022.09.011 (Mol Cell 82(20):3763-3768)	Oct 2022
How human endogenous retroviruses interact with the microbiota in health and disease Dopkins N, O'Mara MM, Singh B, Marston JL, Bendall ML, Nixon DF 10.1016/j.tim.2022.05.011 (Trends Microbiol 30(9):812-815)	Sept 2022

Specific human endogenous retroviruses predict metastatic potential in uveal melanoma Bendall ML, Francis JH, Shoushtari AN, Nixon DF 10.1172/jci.insight.147172 (JCI Insight 7(9))	May 2022
SARS-CoV-2 infection mediates differential expression of human endogenous retroviruses and long interspersed nuclear elements Marston JL, Greenig M, Singh M, Bendall ML, Duarte RR, Feschotte C, Iñiguez LP, Nixon DF 10.1172/jci.insight.147170 (JCI Insight)	Nov 2021
Locus-Specific Characterization of Human Endogenous Retrovirus Expression in Prostate, Breast, and Colon Cancers Steiner MC, Marston JL, Iñiguez LP, Bendall ML, Chiappinelli KB, Nixon DF, Crandall KA 10.1158/0008-5472.CAN-20-3975 (Cancer Res 81(13):3449-3460)	July 2021
Expression of Human Endogenous Retroviruses in Systemic Lupus Erythematosus: Multitomic Integration With Gene Expression Stearrett N, Dawson T, Rahnavard A, Bachali P, Bendall ML, Zeng C, Caricchio R, Pérez-Losada M, Grammer AC, Lipsky PE, Crandall KA 10.3389/fimmu.2021.661437 (Front Immunol 12:661437)	Jan 2021
HAPHPIPE: Haplotype Reconstruction and Phylodynamics for Deep Sequencing of Intra-host Viral Populations Bendall ML, Gibson KM, Steiner MC, Rentia U, Pérez-Losada M, Crandall KA 10.1093/molbev/msaa315 (Mol Biol Evol 38(4):1677-1690)	Apr 2021
Human Endogenous Retrovirus Expression Is Upregulated in the Breast Cancer Microenvironment of HIV Infected Women: A Pilot Study Curty G, Beckerle GA, Iñiguez LP, Furler RL, de Carvalho PS, Marston JL, Champiat S, Heymann JJ, Ormsby CE, Reyes-Terán G, Soares MA, Nixon DF, Bendall ML, Leal FE, de Mulder Rougvié M 10.3389/fonc.2020.553983 (Front Oncol 10:553983)	Jan 2020
Human Endogenous Retrovirus Expression Is Associated with Head and Neck Cancer and Differential Survival Kolbe AR, Bendall ML, Pearson AT, Paul D, Nixon DF, Pérez-Losada M, Crandall KA 10.3390/v12090956 (Viruses 12(9))	Aug 2020
Evaluation of haplotype callers for next-generation sequencing of viruses Eliseev A, Gibson KM, Avdeyev P, Novik D, Bendall ML, Pérez-Losada M, Alexeev N, Crandall KA 10.1016/j.meegid.2020.104277 (Infect Genet Evol 82:104277)	Aug 2020
Validation of Variant Assembly Using HAPHPIPE with Next-Generation Sequence Data from Viruses Gibson KM, Steiner MC, Rentia U, Bendall ML, Pérez-Losada M, Crandall KA 10.3390/v12070758 (Viruses 12(7))	July 2020
A cross-sectional study to characterize local HIV-1 dynamics in Washington, DC using next-generation sequencing Gibson KM, Jair K, Castel AD, Bendall ML, Wilbourn B, Jordan JA, Crandall KA, Pérez-Losada M 10.1038/s41598-020-58410-y (Sci Rep 10(1):1989)	Feb 2020
Transcriptome patterns in hidradenitis suppurativa: support for the role of antimicrobial peptides and interferon pathways in disease pathogenesis Shanmugam VK, Jones D, McNish S, Bendall ML, Crandall KA 10.1111/ced.13959 (Clin Exp Dermatol 44(8):882-892)	Dec 2019

Transcriptomic analysis of human endogenous retroviruses in systemic lupus erythematosus Iñiguez LP, de Mulder Rougvié M, Stearrett N, Jones RB, Ormsby CE, Reyes-Terán G, Crandall KA, Nixon DF, Bendall ML 10.1073/pnas.1907705116 (Proc Natl Acad Sci U S A 116(43):21350-21351)	Oct 2019
Telescope: Characterization of the retrotranscriptome by accurate estimation of transposable element expression Bendall ML, de Mulder M, Iñiguez LP, Lecanda-Sánchez A, Pérez-Losada M, Ostrowski MA, Jones RB, Mulder LCF, Reyes-Terán G, Crandall KA, Ormsby CE, Nixon DF 10.1371/journal.pcbi.1006453 (PLoS Comput Biol 15(9):e1006453)	Sept 2019
Bioconda: sustainable and comprehensive software distribution for the life sciences Grüning B, Dale R, Sjödin A, Chapman BA, Rowe J, Tomkins-Tinch CH, Valieris R, Köster J 10.1038/s41592-018-0046-7 (Nat Methods 15(7):475-476)	July 2018
Benchmark Evaluation of True Single Molecular Sequencing to Determine Cystic Fibrosis Airway Microbiome Diversity Hahn A, Bendall ML, Gibson KM, Chaney H, Sami I, Perez GF, Koumbourlis AC, McCaffrey TA, Freishtat RJ, Crandall KA 10.3389/fmicb.2018.01069 (Front Microbiol 9:1069)	Jan 2018
Genome-wide selective sweeps and gene-specific sweeps in natural bacterial populations Bendall ML, Stevens SL, Chan LK, Malfatti S, Schwientek P, Tremblay J, Schackwitz W, Martin J, Pati A, Bushnell B, Froula J, Kang D, Tringe SG, Bertilsson S, Moran MA, Shade A, Newton RJ, McMahon KD, Malmstrom RR 10.1038/ismej.2015.241 (ISME J 10(7):1589-601)	July 2016
Composition, taxonomy and functional diversity of the oropharynx microbiome in individuals with schizophrenia and controls Castro-Nallar E, Bendall ML, Pérez-Losada M, Sabuncyan S, Severance EG, Dickerson FB, Schroeder JR, Yolken RH, Crandall KA 10.7717/peerj.1140 (PeerJ 3:e1140)	Jan 2015
Dual Transcriptomic Profiling of Host and Microbiota during Health and Disease in Pediatric Asthma Pérez-Losada M, Castro-Nallar E, Bendall ML, Freishtat RJ, Crandall KA 10.1371/journal.pone.0131819 (PLoS One 10(6):e0131819)	Jan 2015
Phylogenomically guided identification of industrially relevant GH1 β-glucosidases through DNA synthesis and nanostructure-initiator mass spectrometry Heins RA, Cheng X, Nath S, Deng K, Bowen BP, Chivian DC, Datta S, Friedland GD, D'Haeseleer P, Wu D, Tran-Gyamfi M, Scullin CS, Singh S, Shi W, Hamilton MG, Bendall ML, Sczyrba A, Thompson J, Feldman T, Guenther JM, Gladden JM, Cheng JF, Adams PD, Rubin EM, Simmons BA, Sale KL, Northen TR, Deutsch S 10.1021/cb500244v (ACS Chem Biol 9(9):2082-91)	Sept 2014
The emergence of lobsters: phylogenetic relationships, morphological evolution and divergence time comparisons of an ancient group (decapoda: achelata, astacidea, glypheidea, polychelida) Bracken-Grissom HD, Ah Yong ST, Wilkinson RD, Feldmann RM, Schweitzer CE, Breinholt JW, Bendall ML, Palero F, Chan TY, Felder DL, Robles R, Chu KH, Tsang LM, Kim D, Martin JW, Crandall KA 10.1093/sysbio/syu008 (Syst Biol 63(4):457-79)	July 2014

Exploring the roles of DNA methylation in the metal-reducing bacterium <i>Shewanella oneidensis</i> MR-1 Bendall ML, Luong K, Wetmore KM, Blow M, Korlach J, Deutschbauer A, Malmstrom RR 10.1128/JB.00935-13 (J Bacteriol 195(21):4966-74)	Nov 2013
Pathoscope: species identification and strain attribution with unassembled sequencing data Francis OE, Bendall ML, Manimaran S, Hong C, Clement NL, Castro-Nallar E, Snell Q, Schaalje GB, Clement MJ, Crandall KA, Johnson WE 10.1101/gr.150151.112 (Genome Res 23(10):1721-9)	Oct 2013
Phylogenetics applied to genotype/phenotype association and selection analyses with sequence data from <i>angptl4</i> in humans Maxwell TJ, Bendall ML, Staples J, Jarvis T, Crandall KA 10.3390/ijms11010370 (Int J Mol Sci 11(1):370-85)	Jan 2010
Phylogenetic relationships and molecular adaptation dynamics of human rhinoviruses Lewis-Rogers N, Bendall ML, Crandall KA 10.1093/molbev/msp009 (Mol Biol Evol 26(5):969-81)	May 2009
Human endogenous retrovirus landscape of melanoma metastases in response to immune checkpoint inhibition Fei T, Singh B, Dopkins N, Marston JL, Francis JH, Nixon DF, Shoushtari AN, Bendall ML (Submitted)	
Single-cell retrotranscriptomics of human germinal center B cells reveals trajectories of cell fate Marston JL, Reyes-Gopar H, Singh B, Greenig M, Dopkins N, Sandoval-Motta S, Lawrence E, O'Mara MM, Fei T, Lin J, Randall KN, Hernandez-Lemus E, Cesarman E, Iñiguez LP, Nixon DF, Bendall ML (In progress)	
Human endogenous retrovirus-mediated transcriptional regulation in the MHC and beyond Greenig M, Bendall ML, Nixon DF, Marston JL* (Under review)	

Current Research Funding

Single cell dark matter in the context of HIV and substance use R01DA063076 (Nixon/Ndhlovu/Corley) · NIH/NIDA · \$2,976,266 TC Y1-5 · 6.00 calendar months · Role: Co-Investigator Submitted in response to RFA-DA-25-016, “Single Cell Opioid Responses in the Context of HIV (SCORCH) Program: Data Mining and Functional Validation (R01 Clinical Trial Not Allowed)”. The application aims to investigate the transcriptional and epigenetic regulatory states of transposable elements at the single cell level in the context of HIV, antiretroviral therapy, and cannabinoid use by applying data mining approaches in multi-omics datasets.	July 2025 – Mar 2030
HOPE – HIV Obstruction by Programmed Epigenetics UM1AI164559 (Ott/Ndhlovu/Valente) · University of California San Francisco/NIH · \$720,376.00 Subaward DC Y1-2 · 3.00 calendar months · Role: Subaward Co-Investigator The HOPE mission is to find a cure for HIV using a novel “block-lock-and-excise” approach—which entails the long-term durable silencing of viral expression towards permanent excision of the latent provirus—leading to the control of the virus in the absence of therapy.	Aug 2021 – Apr 2026

Regulatory Crosstalk Between Human Endogenous Retroviruses, HIV, and EBV, in Lymphoma

May 2021 – Apr 2026

R01CA260691 (Nixon/Cesarman) · NIH/NCI · \$5,904,837.00 TC · 1.08 calendar months · Role: Co-Investigator

- This R01 grant proposal will investigate how these activities intersect and synergize to promote the development of lymphoma in people living with HIV, and how translated HERV products could create neoantigenic targets for immune attack in diffuse large B cell lymphoma (DLBCL).

Past Research Funding

Elucidating the role of endogenous retrovirus expression in primary melanoma and metastasis

July 2021 – June 2024

61500657 (Bendall) · Weill Cornell Medicine (Fund for the Future) · \$300,000.00 TC · 5.76 calendar months · Role: Principal Investigator

- This proposal hypothesizes that locus-specific dysregulation of human endogenous retroviruses (HERV) expression is a significant predictor of metastatic risk in primary uveal melanoma (UM) and metastases associated HERVs modulate anti-tumor immunity and therefore may be harnessed to drive successful immune-based therapy in patients with UM. These hypotheses will be investigated by integrating multiomic data from primary and metastatic UM, leveraging the computational tools and databases that have been developed for analyzing transposable element expression.

The Role of Transposable Elements in Healthy Aging and in Alzheimer's Disease

May 2022 – Aug 2023

R56AG07897001 (Nixon) · NIH/NIA · \$865,823.00 TC · 1.92 calendar months · Role: Co-Investigator

- This R01 proposal aims to investigate the role of human endogenous retroviruses as potent activators of innate immunity and their role in Alzheimer's disease.

Novel approaches for relating genetic variation in endogenous retroviruses to function and disease

Sept 2020 – Aug 2022

R21HG011513 (Nixon/Powell) · NIH · \$150,000.00 DC · 1.20 calendar months · Role: Co-Investigator

- This R21 proposes to use a series of genetic and transcriptomic analyses to infer the etiological and functional role human endogenous retroviruses play in relation to heritable, polygenic disease states.

BELIEVE – Bench to Bed Enhanced Lymphocyte Infusions to Engineer Viral Eradication

July 2016 – June 2022

UM1AI126617 (Nixon) · NIH/NIAID · Role: Co-Investigator

- This proposal hypothesizes that enhancing patients' autologous lymphocytes ex vivo and re-infusing them, along with latency reversing agents, and specific targeting to sites where latent virus resides, could lead to eradication of latently infected cells and potentially a cure for HIV-infected patients.

HIV induced anti-cancer HERV immunity in prostate, breast and colon cancers

Aug 2016 – July 2019

R01CA206488 (Nixon) · NIH/NCI · Role: Graduate Research Assistant

- This proposal hypothesizes that HIV reactivation of HERV's stimulates anti-HERV immunity, which specifically recognizes HERVs also expressed in prostate, colon and breast cancers; and that these HIV induced HERV specific immune responses target HERVs that are expressed in breast, colon or prostate cancer.

Formal Mentorship

Helena Reyes

2021 – present

Postdoctoral Fellow, Feinstein Institutes for Medical Research (2025 – current); PhD Student, Universidad Nacional Autónoma de México (UNAM) (2021 – 2024)

- Advised on Stellarscope software development and computational workflows. First author on Stellarscope manuscript (Cell Reports Methods).

Jez Marston

2020 – present

MD/PhD student, Weill Cornell Medicine

- Advised on HERV cancer and HERV SARS-CoV-2 projects. Published three papers including one first author.

Bhavya Singh, MS

2021 – 2023

Bioinformatician, Weill Cornell Medicine

- Supervised in implementing research workflows for HERV lymphoma and uveal melanoma projects. First author on lymphoma manuscript (iScience).
- Currently a PhD student at the Icahn School of Medicine at Mount Sinai.

Tongyi (Phoebe) Fei, MS

2022 – 2023

Computational Biology Masters student, Weill Cornell Graduate School

- Supervised in implementing research workflows for uveal melanoma projects. First author on melanoma metastasis manuscript (submitted).
- Currently an R&D associate at Sysmex Corporation in Kobe, Japan.

Luis Pedro Iñiguez, PhD

2019 – 2021

Postdoctoral Fellow, Weill Cornell Medicine

- Published four papers including two first-author (PNAS and Front Oncol).
- Currently a Bioinformatician at Centre for Genomic Regulation, Barcelona.

Supervised Trainees

Gislaine Curty, MS

2019 – 2023

Instituto Nacional de Câncer, Rio de Janeiro, Brazil

- Advised on HERV cancer projects. Published four papers with Nixon lab members, including three first-author.

Matthew Greenig

2020 – 2021

Computational Biology MPhil Student, University of Cambridge

- Advised on HERVs in MHC project, manuscript in review. Participant in 2021 Telescope Codeathon.
- Currently a PhD student in Chemistry at University of Cambridge.

Kip Randall

2021

Undergraduate student, San Jose State University

- Co-advised (with Jez Marston) for the Tri-I Gateways to the Lab summer program. Participant in 2021 Telescope Codeathon.

Yesmeen Elgabori

2021

Undergraduate student, Connecticut College

- Co-advised (with Jez Marston) for the Tri-I Gateways to the Lab summer program.

Margaret Steiner Applied mathematics undergraduate, GWU - Advised on viral phylogenetics (HAPHIPE) and HERV cancer projects. Published three papers including one first-author (Cancer Res. 2021). - Currently a PhD student in the Dept. of Human Genetics at University of Chicago.	2019 – 2020
Nathaniel Stearrett, PhD Institute for Biomedical Sciences PhD candidate, GWU Peer advisor on HERVs in SLE work. Published two papers, one first-author (Front Immunol.).	2019 – 2020
Tyson Dawson, PhD Institute for Biomedical Sciences PhD candidate, GWU Peer advisor on HERVs in SLE projects. Published one paper.	2019 – 2020
Keylie Gibson, PhD Computational Biology Institute PhD candidate, GWU - Peer advisor on viral phylogenetics (HAPHIPE), microbiome, and ERV-9 characterization projects. Co-authored five publications including four first-author. - Currently a Bioinformatics Scientist at Integrated DNA Technologies.	2017 – 2020
Gary Hovsepian Pre-Medical Studies undergraduate, GWU - Presented posters at GW Research Day and Frontiers in Retrovirology conference. - Currently a Medical Student at GWU School of Medicine.	2017 – 2018
Mrigank Tiwari Data Science Master's student, GWU - Successfully completed Master's thesis project. - Currently a Senior Chief Engineer at Samsung Electronics, India.	2017
Jingyuan Hu Biology undergraduate, GWU - Co-author of two poster presentations at HIV Dynamics and Evolution conference. - Currently a PhD student at Baylor College of Medicine.	2016 – 2017

Teaching Experience

Lecturer – Thesis research exploration (CMPB 6001) Weill Cornell Graduate School	2020 – 2024
Teaching Assistant – Bioinformatics (BIO465) Brigham Young University	2009 – 2011
Teaching Assistant – Computational Biology (BIO365) Brigham Young University	2009 – 2011

Workshops

Telescope Codeathon – Scientific Organizer Telescope “Dark Matter” Codeathon 2021 Organized and led a four-day codeathon involving 12 biologists and computational scientists. Developed Stellarscope software.	New York, NY Sept 2021
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Telescope intensive course – Lead instructor Endogenous Retroelements in Psychiatry Workshop • Taught two-day workshop introducing RNA-seq and Telescope retrotranscriptome analysis.	London, UK July 2017
PathoScope and PathoStat Tutorial – Instructor Global Biodiversity Genomics Conference • Taught four-hour workshop on microbiome analysis using PathoScope.	Washington, DC Feb 2017
PathoScope and PathoStat Tutorial – Instructor Mid-Atlantic Microbiome Meetup • Taught two-hour workshop on microbiome analysis using PathoScope.	College Park, MD Nov 2016

Software

Stellarscope – Lead Developer Quantification of transposable element expression in single-cells (python)	2023 – present
Telescope – Lead Developer Quantification of transposable element expression using RNA-seq (python)	2015 – present
HAPHPIPE – Lead Developer Haplotype and phylodynamics pipeline for NGS viral assembly (python)	2017 – present
Bioconda – Contributor Sustainable and comprehensive software distribution for the life sciences	2017 – present
PathoStat – Developer Statistical microbiome analysis on metagenomics results (R, Shiny)	2016 – 2018
PathoScope – Developer Strain-level quantification of microbial communities	2013 – 2018
singlecellDB – Lead Developer Analysis system for microbial single cell sequencing (Python/Django, R)	2011 – 2014

Invited Talks

Invited Speaker – Endogenous Retrotransposable Elements and Human Diseases Conference	Mexico Nov 2025
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Honors and Awards

Fund for the Future Department of Medicine, WCM	2021 – 2024
Poster Award Winner Probing Human Disease Using Single-Cell Technologies	Cancun, Mexico May 2022
Biomedical Sciences Fellowship Institute for Biomedical Sciences, GWU	2014 – 2016
Graduate Scholarship Department of Biology, BYU	2009 – 2011
Research Presentation Travel Award Graduate Student Society, BYU	2009

Undergraduate Research Grant

2009

Office of Research and Creative Activities, BYU

Academic Service

Review Editor

Mar 2021 – present

Frontiers in Bioinformatics, Genomic Analysis section

Reviewer

2021

International Journal of Molecular Sciences

LOI Reviewer

2021

Multi-investigator seed grant, Cornell Academic Integration

Reviewer

2021

PLOS Computational Biology

Reviewer

2017

PLOS One

Reviewer

2016

PeerJ